

BACSWN Training Capacity Building Plan

BACSWN — Training & Capacity-Building Plan

Document: BACSWN-TRN-2026-007 **Classification:** CONFIDENTIAL — For Authorized Recipients Only **Prepared for:** Owner & CEO, BACSWN (private delivery partner) / Ministry of Transport & Aviation **Prepared by:** Sky Miles Limited — AI Elevate Division **Version:** 1.0 — June 2026 **Companion to:** Operations & Maintenance Plan (BACSWN-OM-2026-004); Master Framework Schedule (BACSWN-SCHED-2026-003)

1. Purpose & Sovereign Objective

BACSWN is national infrastructure; it must ultimately be **operated and sustained by Bahamians**. This plan builds that capability deliberately — a structured path from partner-led delivery to a Bahamian-staffed watch office, NOC, and field-service organization. **Knowledge transfer is a contracted deliverable, not a courtesy**, and reduces long-term dependence on the partner over the Build-Own-Operate life of the system.

Targets (illustrative, to confirm with the Ministry): Bahamian staffing $\geq 60\%$ by end of program and $\geq 85\%$ within three years of national acceptance, with all front-line watch and NOC roles Bahamian-led.

2. Roles & Competency Framework

Training is **role-based**, each role mapped to a competency standard with assessed certification.

Role	Core competencies
Meteorological watch officers	Aviation weather, SIGMET/ICAO products, hurricane ops, COP/dashboard, decision-making
NOC operators (T1)	Monitoring, alert triage, runbooks, escalation, mesh/autonomy awareness
Field technicians (T2)	Node/radio install & repair, solar/power, antennas, IoT, lightning/grounding, safety
Radar technicians	Raytheon radar operation & maintenance (vendor-certified; ITAR-aware)
Drone / UAS operators	BCAA-licensed flight, survey/SAR/relay missions, airspace deconfliction
Network & software engineers	Mesh protocol, edge AI, hub, OTA, integrations
Security / SecOps	PKI/crypto operations, incident response, pen-test follow-up
Emergency-coordination liaisons	CAP/EDXL, COP, inter-agency procedures with NEMA/Police/Fire/RBDF
Supervisors / managers	Operations management, SLAs, governance, reporting

3. Training Approach

- Train-the-trainer:** build a core of Bahamian instructors early so capability is self-sustaining, not a one-off handover.
- Blended delivery:** classroom + **hands-on at the pilot cluster** + e-learning + on-the-job apprenticeship/shadowing during commissioning.
- Simulation-based:** use the platform's **simulation engine** to run weather, hurricane, and failure scenarios safely — including hub-loss/autonomous-mode and inter-agency activation drills.
- Vendor-delivered specialist training:** Raytheon (radar), Iridium (satellite), NVIDIA/edge, scheduled around install milestones; vendor certificates recorded.
- Competency assessment & certification:** written + practical; certification required before a trainee takes an unsupervised operational role; **recertification** on a defined cadence.

4. Phased Programme (maps to the Schedule)

Phase	Focus
Foundational (M6–M10)	Programme + curricula established; academic partnerships signed; first cohort + train-the-trainer begins; pilot-cluster hands-on

Phase	Focus
Role-specific (M10–M18)	Watch officers, NOC, field techs, radar, drone, security cohorts trained alongside rollout; apprenticeship during commissioning
Operational readiness (M18–M23)	Certification, hurricane-season and inter-agency drills, supervised live operations; Bahamian leads stood up
Sustainment (M23+)	Ongoing intake, refresher/recertification, continuous improvement, progression

5. National Capacity Building & Partnerships

- **Academic & vocational partners:** University of The Bahamas, Bahamas Technical & Vocational Institute (BTVI), the Department of Meteorology, and BCAA — for curricula, accreditation, and a talent pipeline.
- **Pipeline & outreach:** internships, scholarships, and STEM outreach to build a durable local workforce for meteorology, networking, and aviation technology.
- **Localization roadmap:** progressive transfer of front-line, then supervisory, then engineering roles to Bahamian staff, tracked against the §1 targets.
- **Knowledge assets:** SOPs, runbooks (shared with O&M), maintained training materials, and a competency register owned nationally.

6. Governance & Records

- **Training records & competency matrix** maintained per individual and role; tied to operational rostering (O&M Plan §8) — only certified staff hold unsupervised duty.
- **Recertification cadence** and currency tracking (e.g., drone licences, security, watch qualifications).
- **Effectiveness review:** post-drill and post-incident learning feeds curriculum updates; annual review of localization progress with the Ministry.

See also: *Master Framework Schedule (workstream WS-R; gates TRN-1 programme established, TRN-2 operational certification); Operations & Maintenance Plan (BACSWN-OM-2026-004, staffing & rosters).*